Symmetric point sets and the Erdős–Szekeres problem for heptagons

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Abstract

Let $ES(7)$ be the least n such that every n points in general position in the plane contain 7 points in convex position; the Erdős–Szekeres conjecture predicts $ES(7) = 33$. We show, with a computer-assisted proof, that every set of 33 points in general position that has a nontrivial symmetry contains 7 points in convex position. We give an explicit centrally symmetric set of 2^{k-2} points in general position with no k points in convex position for every odd k; for $k = 7$ this is a 2-fold symmetric 32-point set, whose existence was left open by Subercaseaux, Mackey, Qian and Heule. We determine the largest 3-, 4-, 5- and 6-fold rotationally symmetric sets without a convex heptagon (31, 28, 31 and 30 points). The 5-fold bound is a second computer-assisted result: in such sets every convex layer is a single orbit, so a 35-point 5-fold set without a convex heptagon would be seven nested regular pentagons, and none exists. We show that 2-fold and mirror symmetry reach 32, and settle the analogous questions for hexagons; there mirror symmetry stops at 12 points, and the same argument rules out mirror-symmetric extremal sets for octagons.

1 Introduction

Erdős and Szekeres [1] proved that $ES(k)$ is finite and constructed 2^{k-2} points in general position with no k in convex position [2]; they conjectured $ES(k) = 2^{k-2} + 1$. The conjecture holds for $k \le 6$, the case $k = 6$ by a computer search of Szekeres and Peters [3], and $ES(k) \le 2^{k+o(k)}$ by Suk [4]. The first open case is $k = 7$: is every set of 33 points in general position guaranteed a convex heptagon?

A natural place to look for a counterexample is among symmetric configurations, where the search space is much smaller. Subercaseaux, Mackey, Qian and Heule [5] built rotational symmetry into SAT encodings, found symmetric extremal sets for $k = 6$, showed that no 4-fold symmetric 32-point set avoids a convex heptagon, and asked for a 2-fold symmetric one. This note settles the symmetric 33-point question and records several symmetric extremal sets.

Throughout, *general position* means no three points on a line, and a *k-gon* is a set of k points in convex position.

2 Which symmetries are possible

The symmetry group of a finite set fixes its centroid, so it is a cyclic group C_s of rotations or a dihedral group. For a rotation of order s about c, every orbit other than $\{c\}$ has exactly s points and is the vertex set of a regular s-gon.

Lemma 1. *Let S be in general position and invariant under a rotation ρ of order $s \geq 2$ about c. (a) If $s \geq 7$, then S contains a 7-gon. (b) If s is even, then $c \notin S$ and s divides $|S|$. (c) If s is odd, then $|S| \equiv 0$ or $1 \pmod{s}$.*

Proof. (a) An orbit is a convex s-gon. (b) $\rho^{s/2}$ is the half-turn, so $c, p, \rho^{s/2}(p)$ are collinear for every p; hence $c \notin S$ and all orbits have size s. (c) At most the centre is a fixed point. □

In particular a 33-point set in general position can only be symmetric under a reflection or under a rotation of order 3 (without centre): 33 is odd, and $33 \not\equiv 0, 1 \pmod{5}$.

3 Symmetric 33-point sets

Theorem 2 (computer-assisted)**.** *Every set of 33 points in general position in the plane with a nontrivial symmetry contains 7 points in convex position.*

By Lemma 1 there are two cases.

Reflections. A mirror-symmetric 33-point set has exactly one point on the axis. This case was settled earlier in the same project. The encoding is formalized in Lean 4, the 450 hull cases were split into cubes whose CNFs were refuted by CaDiCaL, and every refutation was re-solved and checked from scratch. Details and the trust base are in `results/WRITEUP/MIRROR.md` of the repository.

Rotations by 120°. Here S consists of 11 orbits of size 3. A projective normal form of the labelling (`results/C3-S/soundness.md`) gives a CNF with 168,399 variables and 4,154,921 clauses. Its unsatisfiability was established by splitting on the convex layers and on the orientation bits of symmetric residues into 116,883 certified pieces. For each piece, CaDiCaL produced an LRAT proof that was accepted both by `lrat-check` and by the formally verified checker `cake_lpr` [6]. A reduction by a reflection of the $H6$ strata (Lemma H6M) leaves 910 depth-2 roots. Two independently written composers recompute the CNF hash of every certified piece from the base formula and find all 910 roots covered, with no disagreement (`results/C3FINAL`). As a further check, 2,120 randomly chosen certified pieces were re-solved from scratch; all had identical CNF hashes and both checkers accepted all new proofs (`results/C3SPOT`). The three-fold encoding is tested and audited but, unlike the mirror case, not formalized.

Remark 3. Theorem 2 concerns exactly 33 points. A symmetric set of more than 33 points without a convex heptagon would also refute $ES(7) = 33$. Such a set is excluded by Theorem 2 if it has a 3-fold rotation, or a reflection with a point on the axis, since it then contains a symmetric 33-point subset. It is excluded by [5] if it has a 4-fold rotation, and by Theorem 6 below if it has a 5-fold rotation. Centrally symmetric sets of 34 or more points and mirror-symmetric sets without axis points remain open.

4 Centrally symmetric extremal sets for odd k

Sort points by x-coordinate. An *a-cup* is a sequence of a points whose consecutive slopes increase, an *a-cap* one whose slopes decrease. A set in convex position is the union of its lower chain (a cup) and its upper chain (a cap), which share the leftmost and rightmost points. Let $G(a, b)$ be the classical set of $\binom{a+b-4}{a-2}$ points in general position with no a-cup and no b-cap [1]. The half-turn $h(p) = -p$ reverses the x-order and preserves slopes, so it exchanges cups and caps.

Theorem 4. *For every odd $k \geq 3$ there is a centrally symmetric set of 2^{k-2} points in general position with no k points in convex position.*

Proof. Let $n = k-2$ and $x_i = 2i - n$ for $0 \le i \le n$; these are odd integers with $x_{n-i} = -x_i$. For $i < n/2$ let B_i be a copy of $G(i+2, k-i)$ transformed by $(x, y) \mapsto (\varepsilon x, \varepsilon^2 y)$ and placed at (x_i, x_i^3), and for $i > n/2$ let $B_i = h(B_{n-i})$. The curve $y = x^3$ is h-invariant, so $S = \bigcup_i B_i$ satisfies $h(S) = S$, and $|S| = \sum_i \binom{n}{i} = 2^{k-2}$. Every B_i has no $(i+2)$-cup and no $(k-i)$-cap: for $i > n/2$, B_{n-i} has no $(k-i)$-cup and no $(i+2)$-cap, and h exchanges them.

Three points of the cubic at $x = a, b, c$ are collinear iff $a+b+c = 0$, which is impossible for three odd numbers. Segments inside a cluster have slope of absolute value at most some η that tends to 0 with ε. A segment from B_a to B_b $(a < b)$ has slope within η of $\sigma(a, b) = x_a^2 + x_a x_b + x_b^2 \ge 1$. For $a < b < c$, $\sigma(b, c) - \sigma(a, b) = (x_c - x_a)(x_a + x_b + x_c)$ has absolute value at least 4. Hence for small ε the set is in general position, and consecutive segments between clusters turn according to the sign of $x_a + x_b + x_c$.

Let $P \subseteq S$ be in convex position, with leftmost point $p \in B_l$ and rightmost point $q \in B_r$. If $l = r$, then $|P| \le (l+1) + (k-l-1) - 2 = k-2$. Let $l < r$, with lower chain U and upper chain A.

The slopes of U increase, so after its first segment between clusters it cannot use a segment inside a cluster. Thus U is a cup in B_l (at most $l+1$ points), then single points in clusters $l < c_1 < \cdots < c_t < r$, then q. With $c_0 = l$ and $c_{t+1} = r$, this means $x_{c_{j-1}} + x_{c_j} + x_{c_{j+1}} > 0$ for all j. These sums increase with j, so the condition is equivalent to $x_l + x_{c_1} + x_{c_2} > 0$ (when $t \ge 1$).

The slopes of A decrease, and A must reach $q \in B_r$, so its segments inside clusters all lie in B_r at the end. Thus A is p, then single points in clusters $l < d_1 < \cdots < d_s < r$, then a cap in B_r (at most $k-r-1$ points). With $d_0 = l$ and $d_{s+1} = r$, the decreasing slopes are equivalent to $x_{d_{s-1}} + x_{d_s} + x_r < 0$ (when $s \ge 1$).

If $t, s \ge 1$ and $c_1 \le d_s$, then $0 < x_l + x_{c_1} + x_{c_2} \le x_{d_{s-1}} + x_{d_s} + x_r < 0$, a contradiction. So $d_s < c_1$, the clusters used by U and A between l and r are distinct, and $t + s \le r - l - 1$. Therefore

$$|P| = |U| + |A| - 2 \le (l+1+t+1) + (1+s+k-r-1) - 2 = (k-1) + (t+s) - (r-l-1) \le k-1. \quad \square$$

For $k = 7$ an instance with integer coordinates of absolute value at most 1,250,000 is given in `results/SYM2`. A dynamic program and an independent brute force over all $\binom{32}{7} = 3{,}365{,}856$ seven-point subsets confirm that it has no 7-gon; $k = 9$ (128 points) was checked by the dynamic program. Together with [5, §6.2], this shows that the largest order of a rotational symmetry of a 32-point set without a convex heptagon is 2. For even k the construction fails, because a middle cluster at the centre is collinear with symmetric pairs. Centrally symmetric extremal sets do not exist for $k = 4$ (four centrally symmetric points form a parallelogram) and exist for $k = 6$ [5].

5 Five-fold symmetric sets

Lemma 5. *Let S be in general position and invariant under a rotation ρ of order $m \ge 4$ about a point $c \notin S$, and suppose S has no K points in convex position for some $K \le 2m$. Then every convex layer of S is a single orbit: the orbits can be numbered $O_0, \ldots, O_{t-1}$ so that O_j is the vertex set of $\operatorname{conv}(O_j \cup \cdots \cup O_{t-1})$, and every point of $O_{j'}$ with $j' > j$ lies in the interior of $\operatorname{conv}(O_j)$.*

Proof. The vertex set V of $\operatorname{conv}(S)$ is invariant under ρ, so it is a union of orbits of m points each. It is in convex position, so $|V| < K \le 2m$ and V is one orbit. The other points lie in $\operatorname{conv}(V)$, and by general position not on its boundary. Apply the same argument to $S \setminus V$. $\square$

Theorem 6 (computer-assisted)**.** *Every set of at least 35 points in general position with a rotation of order 5 contains 7 points in convex position.*

Proof outline. By Lemma 1 the set has $5m$ or $5m+1$ points. Removing the centre and whole orbits leaves a 5-fold set S of 35 points without the centre and without a 7-gon. Rotating each orbit by its own small angle keeps the symmetry and all orientations, so we may assume that the centre c is not collinear with two points of S.

The orientations of the 36 points $S \cup \{c\}$ then form a uniform acyclic rank-3 oriented matroid (a CC system) invariant under ρ. We encode it in CNF (`tools/symmetric/symcnf.py`) with:

- the axioms on 4 and 5 points;
- invariance under ρ;
- the orientations of c with each orbit;
- a labelling normal form;
- the absence of a 7-gon, via the 4-point criterion.

Every real configuration gives a model, so unsatisfiability proves the theorem. We ran two normal forms:

- Orbits ordered by angle: 401,464 variables and 7,923,909 clauses, refuted by CaDiCaL in 415 s.
- Orbits ordered by layer, as in Lemma 5, so that S is seven nested regular pentagons: refuted in 54 s.

Both LRAT proofs are accepted by `lrat-check` and by `cake_lpr`. Both normal forms were tested on the real 5-fold sets of 30 and 31 points of Proposition 7, the latter without its centre. Their orientations satisfy the corresponding CNFs, and a deliberately wrong layer order does not (`results/SYM5`). □

For even m the same model has no centre, since the centre is collinear with antipodal pairs. It then does not encode that all segments between antipodal points share their midpoint. Its 4-fold version is satisfiable at 32 points, although no such real set exists [5, §6.2]. So for even-order rotations this model is too weak.

6 The largest symmetric sets without a convex heptagon

Proposition 7. *Let $N(\Gamma)$ be the largest size of a set in general position with symmetry Γ and no 7 points in convex position. Then $N(C_3) = 31$, $N(C_5) = 31$, $N(C_6) = 30$, $N(C_4) = 28$, $N(C_2) \geq 32$ and $N(\mathit{mirror}) \geq 32$. The last two are equalities if $ES(7) = 33$.*

Proof. The lower bounds are explicit sets (`results/SYMMAX`, `results/SYM2`). Each was found by simulated annealing on orbit representatives with exact integer arithmetic: Eisenstein coordinates for 3- and 6-fold symmetry, and coefficients in $\mathbb{Z}[e^{2\pi i/5}]$ with exact sign tests in $\mathbb{Q}(\sqrt{5})$ for 5-fold symmetry. Each was verified by a dynamic program and by brute force over all 7-subsets.

Upper bounds:

- By Lemma 1, a 3-fold set has $3m$ or $3m+1$ points, and with at least 33 points it contains 11 orbits. Theorem 2 excludes 33, and 32 is not a possible size.
- A 6-fold set has $6m$ points and is 3-fold symmetric, so 36 points would contain a 3-fold 33-point set.

- A 4-fold set has $4m$ points; 32 is excluded by [5, §6.2], and larger sets contain a 32-point 4-fold set.
- A 5-fold set has $5m$ or $5m+1$ points; 32 to 34 are not possible sizes, and Theorem 6 excludes 35 or more.
- The other two bounds follow from $ES(7) = 33$ and Lemma 1.

□

The same search reproduces the 16-point hexagon results of [5, §6.1]: 4- and 5-fold examples are found quickly, and none is found for 3-fold symmetry with centre. On 3-fold 33-point sets it never went below three convex heptagons.

7 Mirror symmetry and hexagons

Lemma 8. *Let S be in general position and symmetric under the reflection μ in a line ℓ. Let R be the set of points of S strictly on one side of ℓ, ordered by their projections to ℓ; these are distinct by general position. If $p_1, \dots, p_j \in R$ form a chain in which every middle point is farther from ℓ than the chord of its neighbours, then $\{p_1, \dots, p_j, \mu p_1, \dots, \mu p_j\}$ is in convex position.*

Proof. Traverse $p_1, \dots, p_j, \mu p_j, \dots, \mu p_1$. Along each chain the turns have a common sign because the chain bulges away from ℓ. At p_j the next edge is perpendicular to ℓ, and the turn has the same sign, its cross product being $2\,\mathrm{dist}(p_j, \ell)$ times the projection gap > 0. The same holds at p_1, and the mirror images give the rest. □

Proposition 9. *Let $k = 2j$ be even, and let S be a mirror-symmetric set in general position with no k points in convex position. Then $|S| \le 2\binom{3j-4}{2j-2} + 2$. In particular $|S| \le 12$ for $k = 6$ and $|S| \le 58 < 2^6$ for $k = 8$.*

Proof. In coordinates where the projection to ℓ is the abscissa and the distance to ℓ the height, Lemma 8 says R has no j-cap. R also has no k-cup, because a cup is in convex position. By the cups-caps theorem [1], $|R| \le \binom{k+j-4}{k-2} = \binom{3j-4}{2j-2}$. At most two points lie on ℓ. □

Proposition 10. *For $k = 6$, the largest sets in general position with no convex hexagon and the given symmetry have 16 points for 2-, 4- and 5-fold rotations, 13 points for 3-fold rotations, and 12 points for a reflection.*

Proof. Explicit examples are in `results/SYMMAX/k6`, checked exactly. Since $ES(6) = 17$ [3], 16 is the most possible; the 4- and 5-fold cases also follow from [5]. A 3-fold set has $3m$ or $3m+1$ points. The cases 15 and 16 are impossible: the project's symmetric encoding (`tools/symmetric/symcnf.py`) is UNSAT, with LRAT proofs accepted by `lrat-check` and `cake_lpr`; for 16 this agrees with [5]. The mirror bound is Proposition 9, attained by 5 mirror pairs and 2 points on the axis. □

So mirror symmetry admits no extremal sets for $k = 6$ or $k = 8$, whereas for $k = 7$ it reaches 32 (Proposition 7).

8 Open questions

1. Is there a centrally symmetric set of 34 points in general position without a convex heptagon? Such a set would refute $ES(7) = 33$. A SAT search with 17 antipodal pairs needs a model that includes the centre (see the end of Section 5). With it, the layer structure (centrally symmetric quadrilaterals and hexagons) may help as in the 5-fold case. The mirror case without axis points is open as well.

2. Centrally symmetric sets of 2^{k-2} points without a convex k-gon for even $k \geq 8$ (by Proposition 9, mirror symmetry cannot give them for $k = 8$).

3. A formal verification of the 3-fold encoding, as was done for the mirror case.

References

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